Second Internal Test, May 2024

VIth Semester B. Tech. (F.T.) Degree Programme in Civil Engineering

School of Engineering, CUSAT

19-201-0602 DESIGN OF STEEL STRUCTURES –A batch

(2019 Scheme)

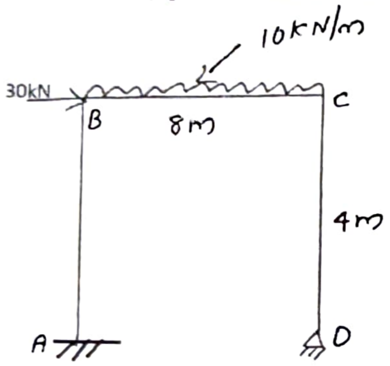
(Assume suitable data wherever necessary,Use of IS Codes: 800-2007, and steel table are permitted in the Examination Hall.)

Duration : 2 hours **Maximum Marks : 30**

1. Apply the provisions in Indian standard codes of practice for the design of various steel structures
2. Design connections, tension and compression members.
3. Design beams and plate girders.
4. Design light gauge steel structures.
5. Design of continuous beams and simple frames using the concept of plastic design.

PART A

(Answer all questions) (2X18=36)

I.	(a)	Find the fully plastic moment for the given frame 	13	CO5
	(b)	List out advantages and disadvantages of welded connection	5	CO2
		OR		
II	(a)	Determine the plastic section modulus Z_{pz} and Z_{py} for ISMC 100 @90.3N/m.	12	CO5
	(b)	State the theorems of plastic analysis	6	CO5
III	(a)	A tie member consisting of an ISA 100 X 75 X 6mm is welded to a 12mm thick gusset plate at site. Design the weld if it is subjected to a factored load of 120kN. Use Fe 410 grade steel.	12	CO2
	(b)	Explain in detail about Type I eccentric connection.	6	CO2
		OR		
IV	(a)	Design a laterally unsupported beam for the following data. Effective span =5m Maximum bending moment = 500kNm. Maximum shear force= 350kN. Use Fe 410 grade steel.	18	CO3